

## Power BI Assignment – DAX for New Insights

### Assignment Tasks

#### 1. Load Data into Power BI (5 Points)

##### ✓ Steps:

- Open Power BI and load the dataset from a CSV file.
- Check that the data is loaded correctly into Power BI.

##### 📌 Expected Output:

- The dataset is successfully loaded into Power BI without any issues.

#### 2. Create a DAX Measure for Total Sales (10 Points)

##### ✓ Steps:

- Create a DAX measure to calculate **Total Sales** (sum of the **Sales** column).
- Display the Total Sales measure in a card visualization.

##### 📌 Expected Output:

- The Total Sales DAX measure is created, and the result is displayed in a card.

#### 3. Create a DAX Measure for Average Sales Per Product (15 Points)

##### ✓ Steps:

- Create a DAX measure to calculate the **Average Sales per Product** (total **Sales** divided by the total number of unique products).
- Display the Average Sales per Product measure in a card visualization.

##### 📌 Expected Output:

- The Average Sales per Product measure is created, and the result is displayed in a card.

#### 4. Create a DAX Measure for Yearly Sales Growth Percentage (15 Points)

##### ✓ Steps:

- Create a DAX measure to calculate the **Yearly Sales Growth Percentage** (percentage change in sales between two consecutive years).
- Display the Yearly Sales Growth in a bar chart for better comparison across years.

##### 📌 Expected Output:

- The Yearly Sales Growth Percentage measure is created and visualized in a bar chart.

#### 5. Create a DAX Measure for Total Profit (15 Points)

##### ✓ Steps:

- Create a DAX measure to calculate **Total Profit** (sum of the **Profit** column).
- Display the Total Profit measure in a table alongside **Product Name** and **Category**.

✦ **Expected Output:**

- The Total Profit DAX measure is created, and the result is displayed in a table.

## 6. Create a DAX Measure for Profit Margin (15 Points)

✓ **Steps:**

- Create a DAX calculated column to calculate the **Profit Margin** for each product ( $\text{Profit} \div \text{Sales}$ ).
- Display the Profit Margin alongside **Product Name** and **Sales** in a table.

✦ **Expected Output:**

- The Profit Margin calculated column is created, and the results are displayed in a table.

## 7. Create Visualizations to Display Insights (20 Points)

✓ **Steps:**

- Create at least two visualizations (e.g., bar chart, pie chart, line graph) to display the following:

1. **Total Sales by Region**

2. **Total Profit by Category**

- Ensure that the visualizations are well-organized, with clear titles and color schemes.

✦ **Expected Output:**

- The visualizations are created and display the total sales by region and total profit by category.

## 8. Publish and Share the Report (5 Points)

✓ **Steps:**

- Share the report with the instructor (in .PBIX file or provide a link to view the report)

✦ **Expected Output:** The report is successfully shared.